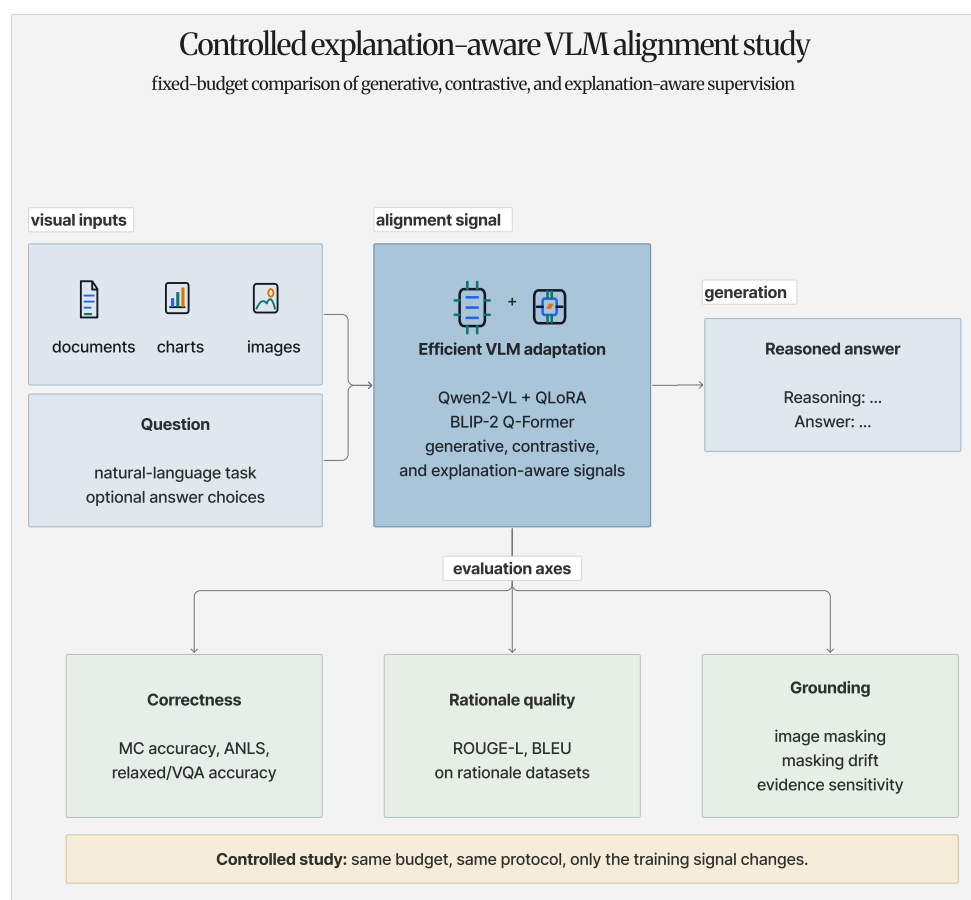


Graphical Abstract

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Highlights

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- The study separates zero-shot, answer-only, rationale-generative, explanation-aware, and BLIP-2 contrastive-enhanced controls under one capped single-GPU protocol.
- On the rationale-bearing datasets, answer-only controls are strong; explanation-aware training is competitive but not automatically best.
- BLIP-2 contrastive-enhanced fine-tuning gives only a small ScienceQA gain over BLIP-2 generative fine-tuning, and the retrieval diagnostic stays near frozen and random controls.
- Evidence-masking drift is reported as evidence sensitivity, so answer gains, rationale quality, and grounding claims are kept separate.

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Abstract

A vision-language model (VLM) is able to read a chart, document, scanned form, or natural image and answer questions about it. But how do we know if the answer is actually derived from the image itself, rather than being a lucky guess on a question based on the model’s remembered language patterns? The correctness of the answer itself is not enough if the reasoning for that answer is not grounded in the visual evidence of the image. Charts and documents are a good example of this problem, because to answer the question correctly, the model has to actually ground its reasoning on specific parts of the image, making it easier to spot when the model is looking at the wrong thing. Although recent work has made progress on this issues, there was no isolation of the training objective as the main variable, with all other factors such as the model, the data, and the budget held constant. Contrastive, generative, and explanation-based methods have all been proposed, but they have not been directly compared under the same conditions, so it is not clear what the objective itself contributed to the final performance. In this paper, we keep the pipeline fixed and change only the training signal, so we can see what the objective does to answer accuracy, explanation quality, and evidence sensitivity when the model, the data, and the compute budget are all held steady. We compare five settings head to head rather than treating them as unrelated systems: BLIP-2 generative fine-tuning, BLIP-2 contrastive-enhanced fine-tuning, Qwen2-VL under answer-only, rationale-generative, and explanation-aware supervision. All the runs use the same resource-constrained single-seed protocol, and the same eval-

uation code as much as possible, across ScienceQA, A-OKVQA, ChartQA, DocVQA, and a VQAv2 subset. The picture that emerges is mixed. On ScienceQA, Qwen2-VL-2B reaches 0.800 accuracy with answer-only supervision, 0.780 in both rationale-generative and length-aware explanation control settings, and 0.765 with fixed-alpha explanation-aware training. On A-OKVQA, the same model reaches 0.830 with answer-only supervision, 0.790 in both rationale-generative and length-aware explanation control training, and 0.795 with fixed-alpha explanation-aware training. ChartQA, DocVQA, and the VQAv2 subset have no gold rationales, so those runs fall back to answer-only controls; they lift the native headline metrics, but they do not test rationale supervision at all. BLIP-2 barely moves: the contrastive objective gives a small gain compared to the generative one on ScienceQA 0.480 vs 0.475, but the retrieval diagnostic stays near frozen and random controls in both cases. Scale is the exception. With a fixed explanation-aware objective on ScienceQA, Qwen2-VL-7B reaches 0.885 while the 2B version reaches 0.765. The masking analysis tells us whether the model relies on the image, not whether its explanation is faithful. Overall, the results do not show a clear win for explanation-aware training. The rationale supervision can help on a single-GPU budget, but the gain is easy to mis-credit to scale or tuning. Hence, it only stands up when answer-only controls, objective balance, model scale, transfer and evidence sensitivity are each reported separately.

Keywords: vision-language models, cross-modal alignment, explanation-aware training, chart and document understanding, faithfulness, parameter-efficient fine-tuning

1. Introduction

Vision-language models (VLMs) have turned into a default tool for reading images, documents, and charts and answering questions about them. The task looks simple from the outside: you hand the model an image and a question, and it should hand back an answer the image actually supports. What counts as evidence, though, depends entirely on the picture. In a document it might be buried in a line of body text, in the table structure, or in one small stamped region of a scanned page. In a chart it might live in an axis label, the relative height of two bars, a colour, or a trend nobody ever wrote down in words. In science and commonsense visual question answering (VQA), it is often not in the image at all, and the model has to fold in background

12 knowledge it already carries. Across every one of these, a confident, correct-
13 looking answer is worth little if the reasoning underneath it is not grounded
14 in the image. Models are very good at sounding right while looking at the
15 wrong thing. So we study the answer and the explanation together, not one
16 or the other.

17 Underneath all of it sits one idea: cross-modal alignment, the job of
18 wiring the visual signal into the language model so the two actually talk.
19 How well that wiring works turns out to hinge on a long list of choices,
20 namely visual resolution, data mixture, connector design, instruction tuning,
21 and the adaptation method [1, 2, 3]. Open model families help, because
22 backbones such as Qwen2-VL, BLIP-2, and LLaVA-OneVision are strong
23 out of the box yet still adaptable on an academic budget [4, 5, 6]. They
24 are not enough on their own, though. Recent document and chart work
25 keeps showing that general visual instruction following falls short on text-
26 heavy, structured images [7, 8, 9, 10]. There is a budget reality too. Most
27 students and small labs cannot fully fine-tune a large multimodal model,
28 full stop. Low-rank methods such as DoRA (weight-decomposed low-rank
29 adaptation) make it realistic to touch only a sliver of the parameters while
30 the base model stays frozen or quantised [11]. We take the same route, using
31 low-rank adaptation (LoRA) and its quantised variant (QLoRA) instead of
32 full-model fine-tuning [12, 13]. The point is never to top a leaderboard; it is
33 to watch what changes when the alignment objective changes and the budget
34 stays real.

35 1.1. Gap Analysis

36 The gap is isolation. A typical paper swaps the model, the data, the
37 training size, and the objective in one move, reports a higher number, and
38 credits the objective. But you cannot actually tell. Maybe the objective
39 helped; maybe the model was simply bigger, or saw cleaner data. Each ob-
40 jective also carries its own blind spot. Contrastive learning is excellent at
41 matching an image to its caption, yet it never directly teaches the model
42 to write an answer or an explanation. Generative learning trains that writ-
43 ing directly, but fluent text is no proof that the pixels were ever consulted.
44 Explanation-aware training goes a step further and supervises the reasoning
45 itself, which is the appealing part; it is also how a model learns to produce a
46 tidy rationale after the fact, decoration rather than cause. And most of the
47 hallucination and faithfulness literature still lives on natural images. Doc-
48 uments and charts get less attention, even though their evidence is more

49 structured and far easier to inspect, which is exactly what makes them a
50 good stress test. We close that gap by changing the alignment objective and
51 holding everything else as steady as we can.

52 1.2. Research Questions

53 The main question is: how should a resource-constrained vision–language
54 model be aligned when the goal is both accurate and faithful visual reasoning?
55 The following research questions make that question measurable.

- 56 1. RQ1: How have vision–language alignment methods evolved for
57 grounded visual reasoning and explanation generation?
- 58 2. RQ2: How does BLIP-2 generative fine-tuning compare with BLIP-2
59 contrastive-enhanced fine-tuning on visual reasoning?
- 60 3. RQ3: Does explanation-aware fine-tuning improve answer accuracy and
61 explanation quality on datasets with gold rationales?
- 62 4. RQ4: How does the same fine-tuning framework perform across docu-
63 ment, chart, science, commonsense, and natural-image domains?
- 64 5. RQ5: How sensitive are the model’s answers to visual evidence under
65 image-masking perturbations?
- 66 6. RQ6: How does the selected explanation-aware QLoRA setup scale
67 from Qwen2-VL-2B to Qwen2-VL-7B?
- 68 7. RQ7: How well do fine-tuned adapters transfer across visual domains
69 and prompt formats?

70 1.3. Problem Statement

71 The problem is cross-modal alignment on the hardware most people actu-
72 ally have: a single GPU. Starting from a pretrained vision–language model,
73 we ask what the objective does to three things at once, namely answer ac-
74 curacy, explanation quality, and how much the model leans on the image.
75 The backbone, the data format, the optimiser, and the evaluation proce-
76 dure are pinned down wherever we can pin them. Inside each comparison,
77 only the training signal moves. The BLIP-2 pair reuses one Q-Former train-
78 ing setup; the Qwen2-VL generative and explanation-aware runs reuse one
79 QLoRA setup. That is what keeps the study tractable instead of ballooning
80 into a full-model fine-tuning project. We are not optimising one benchmark
81 score. The question is narrower and more useful: does the objective move
82 the model in a direction that matters for trustworthy reasoning? A fluent ex-
83 planation with no visual support is a failure here, even when the final answer
84 happens to be right.

85 1.4. *Novelty of this study*

86 The contribution is the control, not a new architecture. Rather than line
87 up unrelated models or recipes against each other, we vary the objective
88 and keep the surrounding setup as close to identical as the budget allows.
89 Faithfulness on structured inputs is treated as a first-class result here, not
90 a couple of cherry-picked figures parked in the discussion. Concretely, the
91 paper contributes the following:

- 92 • A single comparison that puts zero-shot, answer-only, BLIP-2 gener-
93 ative, BLIP-2 contrastive-enhanced, Qwen2-VL rationale-generative,
94 and Qwen2-VL explanation-aware alignment under the same data and
95 optimisation constraints.
- 96 • Evaluation along three separate axes, namely answer quality, explana-
97 tion quality, and evidence sensitivity, across document, chart, science,
98 commonsense, and natural-image tasks.
- 99 • A generate-then-score protocol for multiple-choice reasoning, so the
100 model is judged only after it has produced its own reasoning trace.
- 101 • A direct test of how QLoRA behaves with scale, instead of assuming
102 that full fine-tuning or very large models are the only things worth
103 studying.

104 1.5. *Why this study matters*

105 The usefulness is in the constraints, not in spite of them. The whole
106 study runs on the budget a student or a small lab can actually afford: one
107 GPU, capped data, a single seed. It also refuses the easy move of reporting
108 one number, and keeps answer accuracy, explanation quality, and evidence
109 sensitivity as three separate readings. On documents and charts that separa-
110 tion earns its keep, because there the evidence should trace back to a visible
111 region. The plumbing is reusable as well: every dataset is funnelled into one
112 neutral format, and every experiment is picked from a configuration file. And
113 the ending is honest rather than tidy. Rationale supervision does help on the
114 science and commonsense runs, yet the answer-only controls stay strong, and
115 the BLIP-2 contrastive arm buys only a small answer gain on top of weak
116 retrieval evidence.

117 2. Literature Review

118 This review stays close to the work that touches the present study directly.
119 It is not a full history of vision–language modelling, and does not try to
120 be. We group it by the parts that actually matter here: model design,
121 text-rich images, chart reasoning, faithfulness, datasets, metrics, and low-
122 budget adaptation. Older dataset and metric papers come in only where the
123 experimental protocol leans on them.

124 2.1. Recent vision–language model design

125 No recent VLM rides on a single alignment trick. What it does depends on
126 the backbone, the visual tokenisation, the data mixture, the connector, and
127 the instruction tuning, all interacting. Ablation work makes the case that
128 these knobs are still poorly separated, which is precisely the argument for
129 controlled comparisons [1]. MM1.5 drives it home: optical character recog-
130 nition (OCR) data, synthetic captions, instruction mixtures, and specialised
131 variants can each weigh as much as raw scale [2]. Molmo and PixMo are
132 useful for a different reason, namely open weights and open data that let you
133 actually read the recipe [3]. LLaVA-OneVision shows a single family stretch-
134 ing across image and video once the data mixture is chosen with care [6]. The
135 takeaway is blunt. Change the model, the data, and the budget together,
136 and the objective’s own contribution disappears into the noise.

137 Budget is the other reason open families matter. Qwen2-VL earns its
138 place by handling dynamic resolution, documents, charts, diagrams, and
139 other text-heavy inputs [4]. BLIP-2 earns its place differently: its Q-Former
140 is a convenient spot to bolt on a contrastive image–answer loss while the
141 heavy vision and language components stay frozen [5]. Neither is cited as
142 a leaderboard trophy. They are here as the kind of backbone that makes
143 parameter-efficient experiments possible at all.

144 2.2. Text-rich and document-heavy images

145 Documents are hard for a specific reason: the answer often hides in small
146 text, in the layout, or in local context. Reading and localising the evidence
147 is the task, not naming objects. A recent survey traces how quickly text-rich
148 image understanding has tilted toward multimodal large language models for
149 OCR-heavy and layout-heavy inputs [7]. DocLLM is a clean example, fold-
150 ing layout directly into the generative document-understanding problem [8].

151 OmniDocBench takes the coverage angle instead, scoring diverse PDF pars-
152 ing cases with detailed annotations [14]. Our document task is DocVQA [15].
153 We are after something narrower than document modelling at large: what
154 shifts when the objective changes and the rest of the training path holds
155 still?

156 2.3. Charts and structured visual reasoning

157 Charts break in their own way. Here the model has to read labels, make
158 sense of axes and tick marks, compare quantities, and now and then infer
159 a trend that no one spelled out in words. ChartInstruct and ChartGemma
160 show that instruction tuning on chart images lifts both comprehension and
161 reasoning [9, 10]. ChartMimic adds a useful twist through chart-to-code gen-
162 eration, which probes whether the model grasps the chart’s structure rather
163 than a single answer [16]. ChartMuseum is the sharper warning: on real-
164 world chart reasoning, current large vision-language models (LVLMs) still
165 sit well below humans [17]. MathVista, MMMU-Pro, and MMReason widen
166 the same gap into mathematical, expert-level, and multi-step multimodal rea-
167 soning [18, 19, 20]. That mix is why the design keeps ChartQA, ScienceQA,
168 and A-OKVQA: structured charts, science reasoning, and knowledge-heavy
169 visual reasoning in one sweep [21, 22, 23].

170 2.4. Faithfulness, critique, and efficient adaptation

171 Explanation-aware training carries one obvious risk: the model can learn
172 to write an explanation that reads well while ignoring the image entirely.
173 Recent benchmarks corner this from several sides. MMMU probes expert-
174 level perception and reasoning across many visual formats [24]. MMStar
175 asks the awkward question of whether an example is truly vision-dependent
176 or solvable from language priors alone [25]. HallusionBench zeroes in on
177 where language hallucination meets visual illusion in LVLm answers [26].
178 Hallucination-augmented contrastive learning reframes hallucinated visual
179 text as a representation-learning problem [27]. VISCO and Critic-V push
180 past final-answer accuracy into critique, correction, and error detection [28,
181 29]. Read together, they make one case: measure faithfulness apart from
182 accuracy.

183 Efficient adaptation is the hard constraint we live under. DoRA is the
184 recent example we point to, sharing the same motive: adapt well without
185 paying for full-model fine-tuning [11]. Reason-RFT gestures the other di-
186 rection we care about, toward post-training that explicitly rewards visual

reasoning [30]. Our own work stays with supervised objective comparisons, but the same worry sits behind it. We judge the model as a reasoning system, not merely an answer generator.

Table 1 lays the closest lines of work beside the gap we target. It is not meant as a complete catalogue. The point is narrower: a fixed-budget objective comparison still earns its place even when stronger specialised models already exist.

Table 1: Summary of representative related work and the gap addressed by this study. A check mark indicates that the paper directly targets the property, a cross indicates that it is outside the main scope, and a tilde indicates partial coverage.

Work	Main focus	Generation	Structured VQA	Rationales	Efficient FT	Relation to this study
Recent VLM construction [1, 3, 6, 2]	Backbone and data recipe	✓	~	~	~	Shows why architecture and data must be controlled before claiming an objective-level effect.
Open VLM backbones [4, 5]	Adaptable model families	✓	~	~	~	Provide practical backbone candidates for fixed-budget experiments.
Text-rich document work [7, 8, 14]	Documents and PDF parsing	✓	✓	~	~	Motivates document evaluation, but does not isolate training objectives under one backbone.
Chart reasoning work [9, 10, 17, 16]	Chart understanding	✓	✓	~	~	Shows that chart reasoning remains difficult and needs explicit structured evaluation.
Recent reasoning benchmarks [18, 24, 25, 19, 20]	Harder multi-modal evaluation	~	✓	~	✗	Motivate stricter evaluation for visual, mathematical, and multi-step reasoning.
Faithfulness and critique [26, 27, 28, 29]	Reasoning reliability	~	✓	✓	✗	Supports measuring hallucination, critique, and evidence use separately from accuracy.
DoRA / Reason-RFT [11, 30]	Efficient and reasoning-focused adaptation	~	✗	~	✓	Motivate low-resource adaptation and explicit reasoning-oriented post-training.
This study	Controlled explanation-aware alignment	✓	✓	✓	✓	Isolates objective choice and measures answer, rationale, and evidence-sensitivity outcomes.

One pattern runs through all of it. Early multimodal alignment mostly meant learning a shared image-text representation, or generating a short answer and stopping there. The recent wave pushes much further: harder visual reasoning, instruction tuning, text-rich images, critique, hallucination checks, efficient adaptation. What still goes unmeasured is the objective on its own, namely how much it moves answer quality, rationale quality, and visual evidence use once the model, the data cap, and the compute budget are nailed down. That measurement is the job of the experiments below.

3. Methodology

One rule drives the method: hold the pipeline fixed, change the objective. A dataset goes in and is converted to a shared format. From there the model is loaded, QLoRA adapters are attached if the run needs them, the chosen

objective is trained, and evaluation goes through the same generate-then-score procedure every time. Figure 1 lays out the full pipeline. The design is conservative on purpose. For what we are asking, one clean controlled comparison beats a sprawling leaderboard-style run. Every experiment is declared in a configuration file: the dataset, the backbone, the objective, the adapter settings, the metrics, and the seed, all on the record. One implementation path handles baseline evaluation, fine-tuning, and final evaluation. That is what keeps hidden differences from creeping in between runs.

3.1. Dataset

Five public benchmarks carry the experiments. DocVQA stands in for documents, where the answer usually rides on scanned text, layout, and local visual context [15]. ChartQA covers chart reasoning, where it can turn on a label, an axis, a value, or a comparison [21]. ScienceQA brings multimodal science questions with answer choices and written explanations, which is what makes it usable for rationale supervision [22]. A-OKVQA adds commonsense questions and rationales over natural images [23]. VQAv2 plays the general natural-image control [31]. One caveat there: the completed run draws on the configured `Erland/VQAv2-sample` source, capped at 900 training and 100 evaluation examples, so we report it as a subset result, not a full VQAv2 score. Whatever the source, every dataset is bent into one schema: image, question, optional answer choices, target answer, optional rationale. A dataset with no gold rationale is treated as answer-only, unless a separate rationale experiment is explicitly marked.

Table 2: Benchmarks used in the study. The sizes are rounded public benchmark sizes before filtering by image availability, answer type, or rationale availability. The completed experiments use capped train and evaluation splits so that every run fits the same practical compute budget.

Dataset	Domain	Approx. size	Answer type	Gold rationale	Primary metric
DocVQA [15]	Document	50K	short text span	no	ANLS
ChartQA [21]	Chart	32K	numeric / short text	no	relaxed accuracy
ScienceQA [22]	Science image	21K	multiple choice	yes	accuracy
A-OKVQA [23]	Commonsense image	25K	multiple choice / short text	yes	accuracy
VQAv2 subset [31]	Natural image	900/100 run split	open ended	no	VQA accuracy

3.2. Detailed methodology

Figure 2 shows the model-level setup. Every backbone sits behind one common interface, so the training code never forks into a separate path per

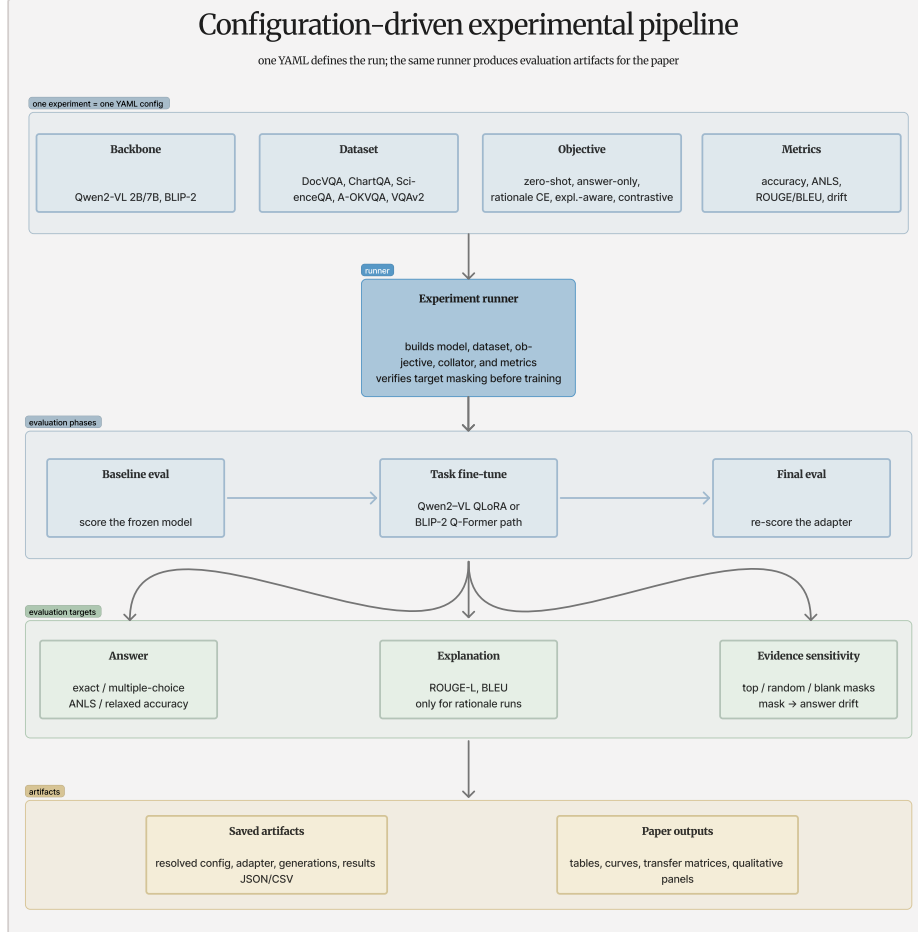


Figure 1: Overall workflow of the proposed explanation-aware alignment study. Datasets are normalised into a shared multimodal example format. Qwen2-VL runs are adapted with QLoRA, BLIP-2 runs train the Q-Former-side alignment path, and each objective is evaluated through answer, rationale, and evidence-sensitivity metrics.

232 model. Qwen2-VL loads in 4-bit precision and picks up rank-16 QLoRA
 233 adapters [12, 13]. On the BLIP-2 contrastive arm, the vision encoder and
 234 the OPT (Open Pre-trained Transformer) language model are frozen solid.
 235 Only the Q-Former-side components, the language projection, and a small
 236 answer-projection head ever update. The baseline variant just evaluates the

Table 3: Input construction examples. The complete token, label, span, and collator traces are saved as pipeline artifacts; this table gives the paper-level view of what each prompt family means.

Prompt family	Raw fields	Prompt seen by model	Supervised target	Supervised spans
Answer-only	image, question, answer	Question: ... plus choices when available	Answer: <answer>.	answer tokens only
Rationale-generative	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	full target under uniform CE
Explanation-aware	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	rationale and answer spans weighted separately
Answer-only fall-back	image, question, answer, no gold rationale	Question: ...	Answer: <answer>.	answer tokens only

237 model before any task fine-tuning. The contrastive-enhanced variant bolts on
 238 an auxiliary image–answer term: it tugs the pooled Q-Former image repre-
 239 sentation toward the correct answer and away from the others in the batch.
 240 The generative variant trains the model to produce the answer from the
 241 image and question. The explanation-aware variant trains it to produce a
 242 rationale first, then the answer. So the variants differ in one thing only, the
 243 supervision signal, and everything else inside a comparison is held as close
 244 as it can be.

245 In generative training, prompt and target live in one concatenated se-
 246 quence, but the loss only ever touches the target tokens. That detail is easy
 247 to get wrong, because multimodal processors slip image placeholder tokens in
 248 front of the text. Count the prompt length by hand and the supervised span
 249 quietly slides off position. The implementation sidesteps it by measuring
 250 prompt length with the very same processor that encodes the full sequence.
 251 The contrastive objective reuses those span tags, pooling answer tokens alone
 252 into the InfoNCE (information noise-contrastive estimation) text representa-
 253 tion [32]. Rationale tokens still receive the generative loss; they simply are
 254 not treated as the contrastive positive. For the explanation-aware objective,
 255 the target splits in two, an explanation span and an answer span. Equation 1
 256 weights the two spans with α . When $\alpha = 1$, only the answer span is weighted,
 257 although the target sequence can still contain the gold rationale before the
 258 answer. A clean answer-only control is therefore reported separately, using
 259 a target template that contains only **Answer:**. When $\alpha = 0$, only the ex-
 260 planation span is supervised. Intermediate values test whether answer and
 261 rationale supervision help each other or compete.

$$\mathcal{L} = \alpha \mathcal{L}_{\text{answer}} + (1 - \alpha) \mathcal{L}_{\text{explanation}} \quad (1)$$

262 where $\mathcal{L}_{\text{answer}}$ and $\mathcal{L}_{\text{explanation}}$ are the masked cross-entropy losses over the

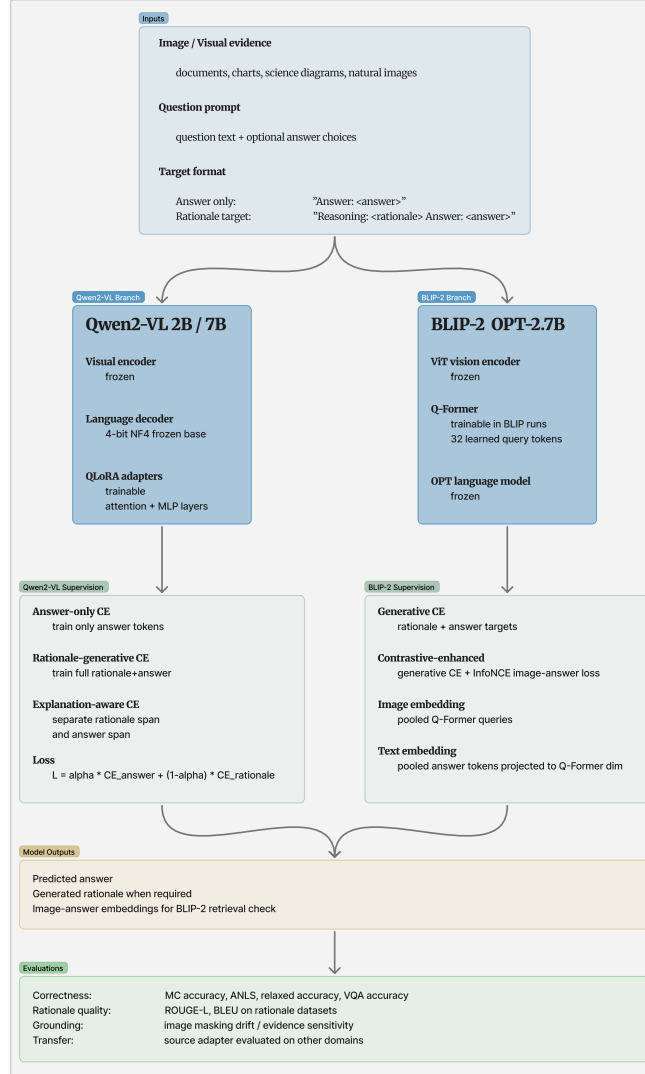


Figure 2: Model-level architecture for explanation-aware cross-modal alignment. Visual evidence and the question prompt are encoded into a multimodal token stream. The Qwen2-VL path uses trainable QLoRA adapters and separate rationale and answer supervision spans. The BLIP-2 path keeps the vision tower and OPT language model frozen and trains the Q-Former-side generative or contrastive alignment components.

263 answer and explanation token spans, and α controls their relative weight.
 264 There are two alpha modes. Fixed-alpha mode treats α as a dial we set
 265 by configuration, an ablation knob. Length-aware mode does something

different: it derives the effective answer weight straight from the supervised token counts:

$$\alpha_{\text{eff}} = \frac{\lambda n_{\text{answer}}}{\lambda n_{\text{answer}} + n_{\text{explanation}}}, \quad (2)$$

where n_{answer} and $n_{\text{explanation}}$ are the answer and explanation token counts after collation, and λ is the answer-weight multiplier. With $\lambda = 1$, this collapses to plain uniform token-level cross-entropy over the rationale-plus-answer target. So the fixed-alpha sweep is the ablation; the length-aware run is the audit control for the natural, token-weighted setting.

Multiple-choice datasets are scored in two steps at inference. First the model writes a short reasoning trace from the image and question. Then, in the context of that trace, each candidate answer gets a length-normalised conditional likelihood, and the highest score wins. The two-step design is there to avoid brittle string parsing and option-label copying. Open-ended datasets are simpler: the generated answer string is scored with the dataset’s native metric. Whenever the model produces a rationale, we save it, and those saved rationales feed the text-quality and faithfulness analysis.

3.3. Evaluation metrics

Evaluation splits into three parts: answer quality, rationale quality, and evidence sensitivity. Answer quality always uses the benchmark’s own native metric. When the tables say “headline metric,” they mean exactly that native primary metric for the target dataset, not some universal number to be read straight across datasets. For ScienceQA and A-OKVQA that metric is multiple-choice accuracy. DocVQA uses average normalised Levenshtein similarity (ANLS), per the benchmark convention [15]. ChartQA uses relaxed accuracy on numerical answers [21]. The VQAv2 subset uses VQA consensus accuracy wherever annotator answers exist [33, 31]. Rationale quality leans on BLEU and ROUGE-L whenever gold rationales are available [34, 35]. Neither is a perfect judge, but together they give a basic lexical and sequence-level check on the explanation. Evidence sensitivity asks a blunter question: does the answer’s likelihood move when we mask part of the image? This borrows the old idea of occlusion sensitivity analysis [36]. The specific measure is evidence-masking consistency, written out in Equation 3. The qualitative diagnostic then lines up the high-drift regions against whatever visual evidence the generated explanation actually names, when that comparison is possible.

$$\text{drift}(r) = s(a \mid I) - s(a \mid I \setminus r) \quad (3)$$

where $s(a \mid I)$ is the length-normalised log-likelihood of the gold answer a given image I , and $I \setminus r$ is the same image with region r occluded. A large positive drift says the masked region mattered. A drift near zero says the opposite: the answer probably leaned on language priors, a shortcut, or non-visual context. The absolute scale does not travel across datasets, since answer length, prompt family, image resolution, and task format all bend the likelihood scores. The split it captures is the familiar one between a useful explanation and a faithful one. It also speaks to recent hallucination and critique work, where a model happily describes or relies on visual content that simply is not there [26, 27, 28, 29]. That is why evidence sensitivity sits next to answer accuracy here, rather than being parked as a post-hoc example.

3.4. Experimental settings

Every variant runs under the same fixed-budget protocol, and any deviation from it is written into the experiment ledger. The Qwen2-VL runs share one recipe: AdamW, a learning rate of 2×10^{-4} , cosine decay, three percent warm-up, weight decay of 0.01, bf16 compute, and an effective batch size of 32 reached through gradient accumulation. The base model loads under 4-bit NormalFloat (NF4) quantisation. Unless a result table says otherwise, the QLoRA adapters use rank 16 and scaling factor 32 [12, 13]. The workhorse backbone is Qwen2-VL-2B-Instruct, strong enough for the target tasks yet small enough to fit the budget [4]. For BLIP-2 we use Salesforce BLIP-2 OPT-2.7B, vision tower and OPT language model frozen, training only the Q-Former-side alignment components [5]. The scale ablation reuses the same QLoRA recipe on Qwen2-VL-7B. Each run logs its dataset split, filtered count, seed, objective, adapter configuration, and metrics. The pipeline also writes a split-and-leakage audit from the selected train and evaluation identifiers, wherever the dataset exposes stable IDs. For the capped datasets in the final artifact set, that overlap count comes back at zero. DocVQA is the one non-standard split: the loaded dataset only exposes validation and test, so the controlled run trains on `validation[:80%]` and evaluates on `validation[80%:]`, with native-ID overlap checking. The audit shows zero native-ID overlap, but it cannot prove document-level disjointness past the IDs the loader exposes. So we treat the DocVQA number as a capped internal adaptation result, not a public-test benchmark claim. We also attach

335 approximate uncertainty estimates to the accuracy-style headline metrics.
 336 With most evaluation caps at 200 examples and only a few seeds in the bud-
 337 get, a gap like 0.790 versus 0.795 is an observed difference, nothing more.
 338 Table 4 lists the default configuration.

Table 4: Default Qwen2-VL QLoRA fine-tuning configuration. BLIP-2 contrastive runs use the separately reported frozen-vision/frozen-LLM Q-Former training setup. Any run that changes these values is reported separately in the experiment ledger and in the Results section.

Training Configuration	
Adapter	QLoRA (4-bit NF4, rank 16, scaling 32)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	2×10^{-4} (cosine, 3% warm-up)
Weight decay	0.01
Effective batch size	32 with gradient accumulation
Precision	bf16 compute
Primary seed policy	one seed per complete configuration

339 3.5. Experiment ledger

340 Before any scores, Table 5 lays out the comparisons. The line that mat-
 341 ters is rationale availability. ScienceQA and A-OKVQA can carry rationale-
 342 generative and explanation-aware supervision, because they ship gold ratio-
 343 nales. ChartQA, DocVQA, and VQAv2 stay on answer-only fallback, plus
 344 the transfer and domain runs, until some future experiment adds rationale
 345 supervision of their own.

Table 5: Main experiment ledger. The full generated ledger includes exact caps, seeds, prompt variants, alpha mode, and configuration hashes.

Comparison	Runs included	Use in the paper
BLIP-2 objective pair	BLIP-2 OPT-2.7B on ScienceQA: generative and generative+contrastive.	Tests whether the auxiliary contrastive objective adds anything beyond BLIP-2 generative fine-tuning.
ScienceQA controls	Qwen2-VL-2B on ScienceQA: answer-only, rationale+answer CE, fixed- α explanation-aware sweep, and length-aware control.	Separates answer supervision, rationale supervision, and alpha weighting on a gold-rationale dataset.
A-OKVQA controls	Qwen2-VL-2B on A-OKVQA: answer-only, rationale+answer CE, fixed- α explanation-aware, and length-aware control.	Repeats the rationale-supervision comparison on a second gold-rationale domain.
Fallback domains	Qwen2-VL-2B on ChartQA, DocVQA, and VQAv2 with answer-only fallback prompts.	Reports in-domain adaptation and transfer for datasets without gold rationales; these rows are not used as explanation-aware evidence.
Scale	Qwen2-VL-2B and Qwen2-VL-7B on ScienceQA with the same fixed explanation-aware QLoRA setting.	Checks whether the selected setup changes when the backbone scale increases.
Transfer	Qwen2-VL-2B source adapters evaluated on target-domain prompts across all datasets.	Tests cross-domain behaviour separately from the official in-domain result table.
Masking	Selected Qwen2-VL runs with image-region perturbation and saved visual examples.	Measures evidence sensitivity beside answer accuracy and rationale quality.

346 4. Results

347 The results track the comparison types from the ledger, in order. The
348 BLIP-2 objective pair comes first, testing whether the auxiliary contrastive
349 term earns its place beyond generative fine-tuning. Then the ScienceQA
350 and A-OKVQA runs pull answer-only, rationale-generative, and explanation-
351 aware supervision apart on the two datasets that have gold rationales. The
352 chart, document, and VQAv2 runs follow, showing in-domain adaptation
353 where no gold rationales exist. After that come masking-based evidence sen-
354 sitivity, the Qwen2-VL scale check, and cross-domain transfer. One reminder
355 throughout: every score comes from the capped single-seed protocol above,
356 so they are controlled framework results, not leaderboard numbers.

357 4.1. Objective-level comparison

358 Table 6 holds the direct objective comparisons. Take the BLIP-2 Sci-
359 enceQA pair first: both trained variants clear the 0.445 frozen baseline, the
360 contrastive-enhanced run landing at 0.480 and the generative control at 0.475.
361 That gap is tiny, and it sits inside the approximate uncertainty intervals in
362 Table 7. Retrieval tells the same cautious story: the contrastive-enhanced
363 model posts recall@1 (R@1) of 0.010, R@5 of 0.020, R@10 of 0.030, and a
364 mean reciprocal rank (MRR) of 0.0307. Qwen2-VL-2B is stronger across
365 the board. Rationale-generative fine-tuning reaches 0.780 on ScienceQA and
366 0.790 on A-OKVQA. On A-OKVQA the explanation-aware run reaches 0.795
367 accuracy, 0.337 ROUGE-L, and 0.087 BLEU, numerically a hair above the
368 generative control, which we read as an observed single-run difference rather
369 than a settled win. The honest surprise is the answer-only controls: 0.800
370 on ScienceQA and 0.830 on A-OKVQA, both strong. The zero-shot col-
371 umn, meanwhile, is one shared backbone-dataset reference for orientation,
372 not a fresh prompt-family baseline per row. Back on ScienceQA, the fixed
373 $\alpha = 0.50$ explanation-aware run reaches 0.765. The length-aware control
374 reaches 0.780, level with the rationale-generative control but still under the
375 answer-only number.

Table 6: Objective-level results under the controlled protocol. The zero-shot reference is the shared pre-fine-tuning score for the same backbone and dataset group, not a separately recomputed frozen score for every prompt family. Accuracy is multiple-choice accuracy for ScienceQA and A-OKVQA.

Backbone	Dataset	Objective	Zero-shot ref.	Final	ROUGE-L	BLEU
BLIP-2 OPT-2.7B	ScienceQA	Generative	0.445	0.475	–	–
BLIP-2 OPT-2.7B	ScienceQA	Contrastive-enhanced	0.445	0.480	–	–
Qwen2-VL-2B	ScienceQA	Answer-only	0.400	0.800	–	–
Qwen2-VL-2B	ScienceQA	Rationale+answer CE	0.400	0.780	0.693	0.559
Qwen2-VL-2B	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	0.643	0.493
Qwen2-VL-2B	ScienceQA	Explanation-aware, length-aware	0.400	0.780	0.696	0.563
Qwen2-VL-2B	A-OKVQA	Answer-only	0.355	0.830	–	–
Qwen2-VL-2B	A-OKVQA	Rationale+answer CE	0.355	0.790	0.329	0.083
Qwen2-VL-2B	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	0.337	0.087
Qwen2-VL-2B	A-OKVQA	Explanation-aware, length-aware	0.355	0.790	0.330	0.083

Table 7: Approximate uncertainty for selected accuracy-style headline metrics. These intervals use the generated normal-approximation standard errors for capped evaluation sets, so they are a caution about small observed differences rather than a replacement for multi-seed testing.

Run	Metric	Eval cap	Score	Approx. 95% CI
BLIP-2 generative ScienceQA	MC accuracy	200	0.475	0.406–0.544
BLIP-2 contrastive ScienceQA	MC accuracy	200	0.480	0.411–0.549
ScienceQA answer-only	MC accuracy	200	0.800	0.745–0.855
ScienceQA rationale+answer CE	MC accuracy	200	0.780	0.723–0.837
ScienceQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.765	0.706–0.824
ScienceQA explanation-aware, length-aware	MC accuracy	200	0.780	0.723–0.837
A-OKVQA answer-only	MC accuracy	200	0.830	0.778–0.882
A-OKVQA rationale+answer CE	MC accuracy	200	0.790	0.734–0.847
A-OKVQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.795	0.739–0.851
Qwen2-VL-7B ScienceQA, $\alpha = 0.50$	MC accuracy	200	0.885	0.841–0.929

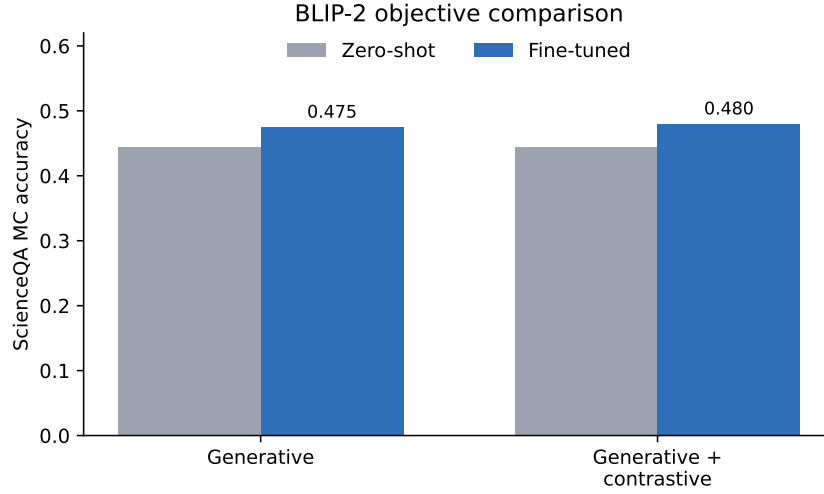


Figure 3: BLIP-2 generative and contrastive-enhanced ScienceQA results. Both runs improve over the frozen baseline, with only a small observed gain from the auxiliary contrastive term.

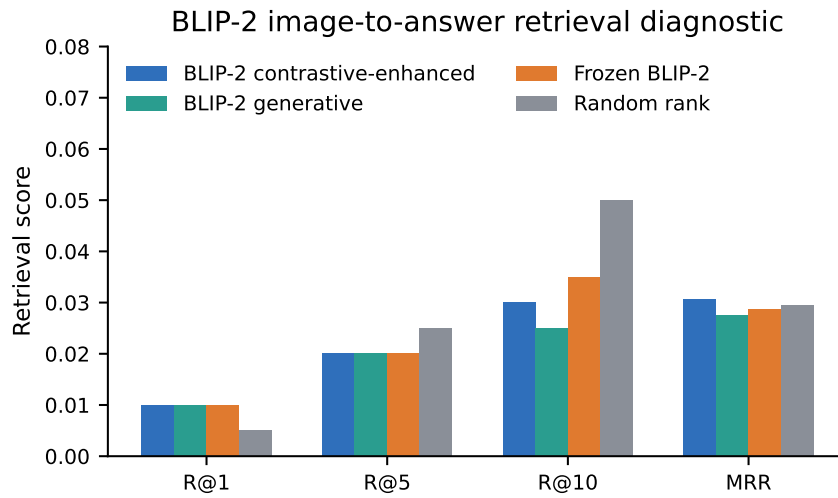


Figure 4: BLIP-2 image-to-answer retrieval diagnostic. The analysis reports frozen BLIP-2, BLIP-2 generative, BLIP-2 contrastive-enhanced, and random-rank controls so the contrastive branch is not interpreted without a reference point.

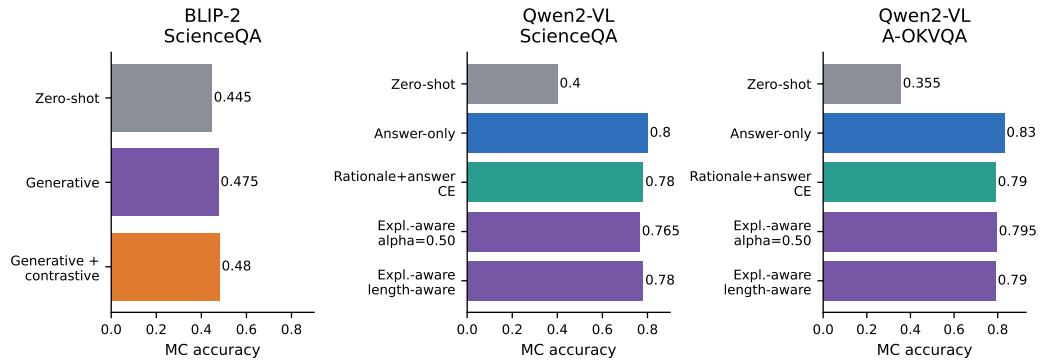


Figure 5: Controlled strategy comparison for BLIP-2 and Qwen2-VL. The figure shows zero-shot, answer-only, rationale-generative, explanation-aware, and contrastive variants where those controls are available.

376 *4.2. Effect of explanation-aware training*

377 Table 8 and Figure 6 trace the ScienceQA α -sweep, where α is the answer-
 378 span weight from Equation 1. Fine-tuning alone lifts accuracy from the
 379 0.400 zero-shot reference into the 0.690–0.780 band. The sweet spot is low:
 380 $\alpha = 0.10$, at 0.780. Push α toward the answer-heavy end and accuracy slides,
 381 to 0.690 at $\alpha = 0.75$ and 0.695 at $\alpha = 1.00$. The explanation metrics move
 382 in lockstep. ROUGE-L and BLEU also peak at $\alpha = 0.10$ (0.695 and 0.566)
 383 and decay as α climbs. At $\alpha = 1.00$ only the answer span is supervised
 384 inside a rationale-shaped target, so there are no rationale metrics to report.
 385 In accuracy, the $\alpha = 0.10$ run ties the rationale-generative and length-aware
 386 controls. One trap worth flagging: the $\alpha = 1.00$ row is not the true answer-
 387 only control, because the model still sees a rationale-shaped template even
 388 though the rationale span goes unsupervised. The genuine answer-only Sci-
 389 enceQA control, on its own prompt family, reaches 0.800. And all of this
 390 rests on one seed, 2,000 training and 200 evaluation examples, so these are
 391 observed effects, not final statistical proof.

Table 8: Explanation-aware fixed- α sweep on ScienceQA (Qwen2-VL-2B, QLoRA, 2,000 train / 200 eval, single seed). The coefficient α weights the answer span in Equation 1: $\alpha = 1$ is answer-span-only within the rationale target and $\alpha = 0$ is explanation-span-only. The zero-shot and rationale-generative rows are references. Random-choice accuracy is 0.354. The best fixed α is shown in bold.

Variant (α)	MC accuracy	ROUGE-L	BLEU
Zero-shot (frozen)	0.400	–	–
Rationale+answer CE	0.780	0.693	0.559
$\alpha = 0.00$ (explanation-only)	0.725	0.417	0.299
$\alpha = 0.10$	0.780	0.695	0.566
$\alpha = 0.25$	0.775	0.686	0.548
$\alpha = 0.50$	0.765	0.643	0.493
$\alpha = 0.75$	0.690	0.604	0.443
$\alpha = 1.00$ (answer-span only)	0.695	–	–

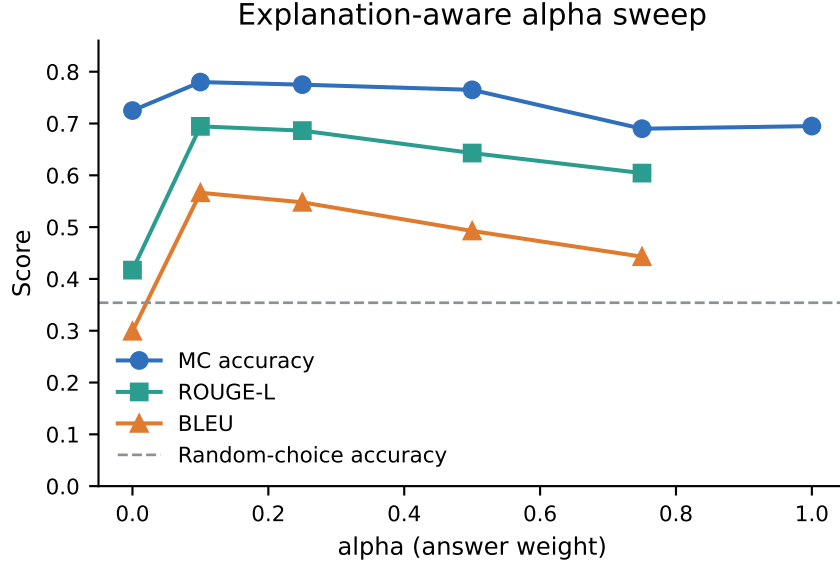


Figure 6: Fixed-alpha explanation-aware sweep on ScienceQA. Answer accuracy peaks at the low-answer-weight setting and declines toward both extremes inside the rationale-plus-answer target. Explanation quality (ROUGE-L, BLEU) peaks at the same point and falls as α increases. Series are distinguished by marker shape and line style; the dashed grey line marks random-choice accuracy. Single seed.

4.3. Domain-level behaviour

Table 9 and Figure 7 gather the selected domain runs from the final matrix. For ScienceQA the row uses the fixed $\alpha = 0.50$ explanation-aware run, so it lines up with the in-domain figure; the stronger $\alpha = 0.10$ and length-aware settings live in the sweep table above. The biggest jumps land on DocVQA, VQAv2, ChartQA, and A-OKVQA. A-OKVQA climbs from 0.355 to 0.795 accuracy against the reported zero-shot reference. The VQAv2 subset goes from 0.350 to 0.730 VQA accuracy. ChartQA reaches 0.655 relaxed accuracy, exact match 0.410. DocVQA is the headline, at 0.905 ANLS with exact match at 0.855. That number is strong for a capped internal run, but it deserves caution on three counts: the qualitative samples are short extractive answers, the setup is answer-only fallback rather than gold rationales, and the split audit cannot prove document-level disjointness past the native IDs it can see.

Table 9: Domain-level results for the completed Qwen2-VL-2B runs. Each domain uses its native headline metric: multiple-choice accuracy for ScienceQA and A-OKVQA, relaxed accuracy for ChartQA, ANLS for DocVQA, and VQA accuracy for the VQAv2 subset.

Domain	Dataset	Objective	Zero-shot ref.	Final	Secondary metric
Science	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	ROUGE-L 0.643, BLEU 0.493
Commonsense	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	ROUGE-L 0.337, BLEU 0.087
Charts	ChartQA	Answer-only fallback	0.335	0.655	Exact match 0.410
Documents	DocVQA	Answer-only fallback	0.349	0.905	Exact match 0.855
Natural images	VQAv2 subset	Answer-only fallback	0.350	0.730	Exact match 0.350 $\rightarrow$ 0.730

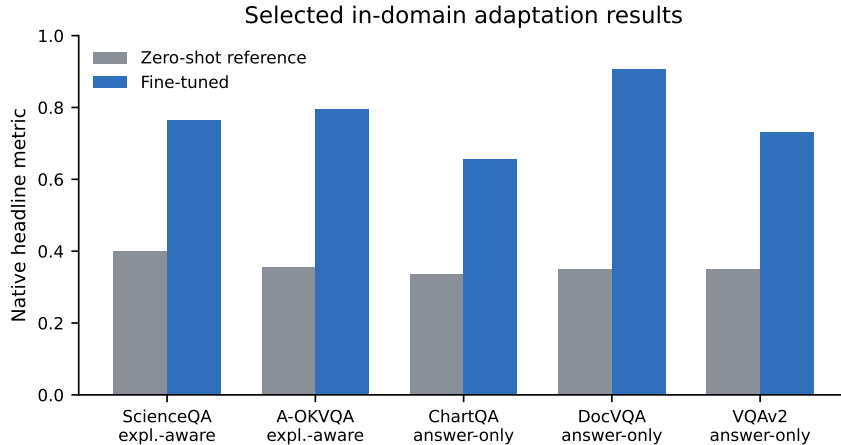


Figure 7: In-domain adaptation results for the selected completed Qwen2-VL-2B domain runs. The comparison mixes rationale-supervised ScienceQA/A-OKVQA rows with answer-only fallback rows for domains without gold rationales, so it should not be read as an explanation-aware effect across all domains.

4.4. Faithfulness and hallucination analysis

Table 10 and Figure 8 carry the evidence-masking drift check. Each number is a mean over 30 evaluation examples. The bigger and more positive it is, the more masking a region costs the gold-answer likelihood, which means the model was relying on what we hid. The drift piles up on the structured tasks: DocVQA at 0.375, ChartQA at 0.224, VQAv2 at 0.111. ScienceQA and A-OKVQA usually sit far lower, with one loud exception, the ScienceQA answer-only adapter at 0.0898. That low ScienceQA drift is not surprising; the questions come loaded with language priors and explicit answer choices, the visual evidence is diffuse, and a coarse mask can miss the single cue the model actually used. Explanation supervision does not move drift in any clean way here. At $\alpha = 0.10$ it is 0.0223, at $\alpha = 0.50$ it is 0.0197, and answer-span-only $\alpha = 1.00$ gives 0.0149, despite that run producing no supervised

419 rationale at all. On A-OKVQA the explanation-aware run drifts a touch
 420 more than the generative control, 0.0271 against 0.0245, but the intervals
 421 around both are wide. Read plainly, the masking result backs a domain-
 422 sensitivity story, not the sweeping claim that explanation-aware training by
 423 itself buys faithfulness. We report no manual hallucination rate, because we
 424 never ran a fixed annotation protocol for hallucination categories.

Table 10: Evidence-masking drift by completed run. Higher values indicate stronger sensitivity to masked visual evidence. The intervals are approximate 95% intervals from the generated masking artifact.

Run	Mean drift	Approx. 95% CI	Examples
ScienceQA answer-only	0.0898	0.0197–0.1830	30
ScienceQA rationale+answer CE	0.0217	-0.0100–0.0582	30
ScienceQA $\alpha = 0.10$	0.0223	-0.0080–0.0568	30
ScienceQA $\alpha = 0.50$	0.0197	-0.0119–0.0552	30
ScienceQA answer-span-only $\alpha = 1.00$	0.0149	-0.0274–0.0638	30
A-OKVQA rationale+answer CE	0.0245	-0.0236–0.0726	30
A-OKVQA explanation-aware	0.0271	-0.0075–0.0627	30
ChartQA answer-only fallback	0.2237	0.1419–0.3140	30
DocVQA answer-only fallback	0.3749	0.2770–0.4960	30
VQAv2 answer-only fallback	0.1109	0.0274–0.2408	30
Qwen2-VL-7B ScienceQA $\alpha = 0.50$	0.0114	-0.0059–0.0260	30

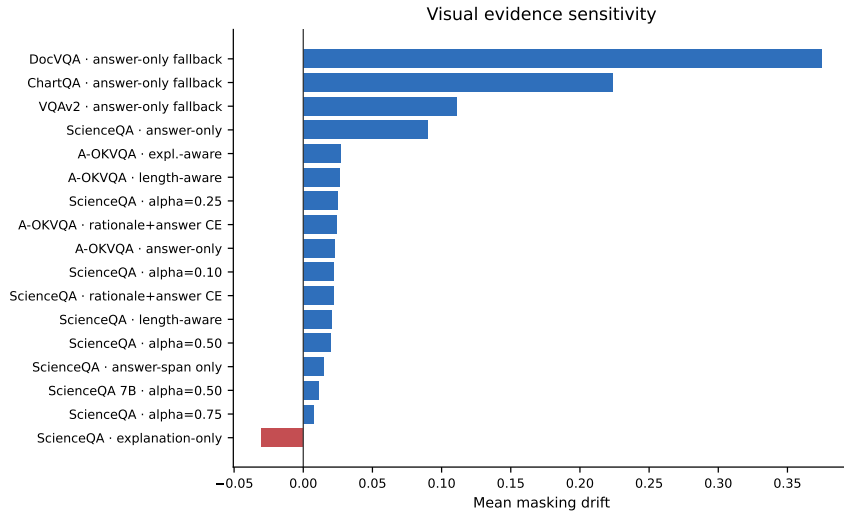


Figure 8: Mean evidence-masking drift across completed runs. Structured document and chart runs show much stronger visual sensitivity than the rationale-rich ScienceQA and A-OKVQA runs, but stronger drift does not automatically imply better answer accuracy.

ChartQA masking diagnostic: top-drift mask hurts the answer more than a random mask

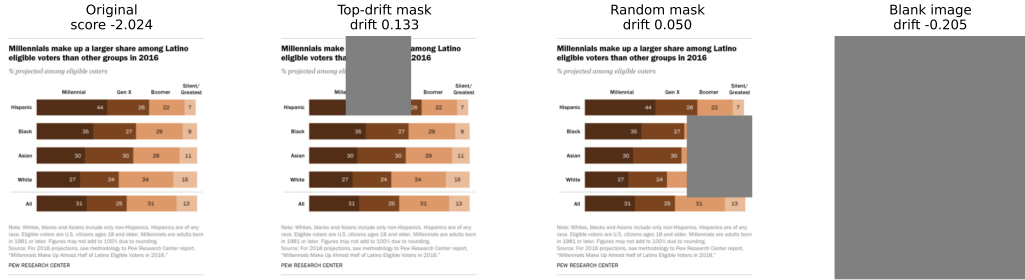


Figure 9: Example masking diagnostic for a ChartQA answer-only fallback run. The pipeline saves the original image, the top-drift mask, a random-region mask, and a blank-image control, together with the full-image and masked answer scores. This figure is used as an evidence-sensitivity example, not as proof that a generated rationale is faithful.

4.5. Efficiency and scale trade-offs

Table 11 and Figure 10 put Qwen2-VL-2B and Qwen2-VL-7B side by side on the same ScienceQA explanation-aware setting ($\alpha = 0.50$). Scaling up pays. The 7B run lifts accuracy from 0.765 to 0.885. ROUGE-L rises from 0.643 to 0.738, BLEU from 0.493 to 0.637. It is the single largest gain anywhere in the completed set. Both runs still fit under QLoRA. What the logs do not give us is a stable wall-clock or peak-memory figure, so this stands as a model-scale result, not a precise efficiency one.

Table 11: Qwen2-VL scale comparison on ScienceQA with the same explanation-aware objective ($\alpha = 0.50$).

Model	MC accuracy	ROUGE-L	BLEU
Qwen2-VL-2B	0.765	0.643	0.493
Qwen2-VL-7B	0.885	0.738	0.637

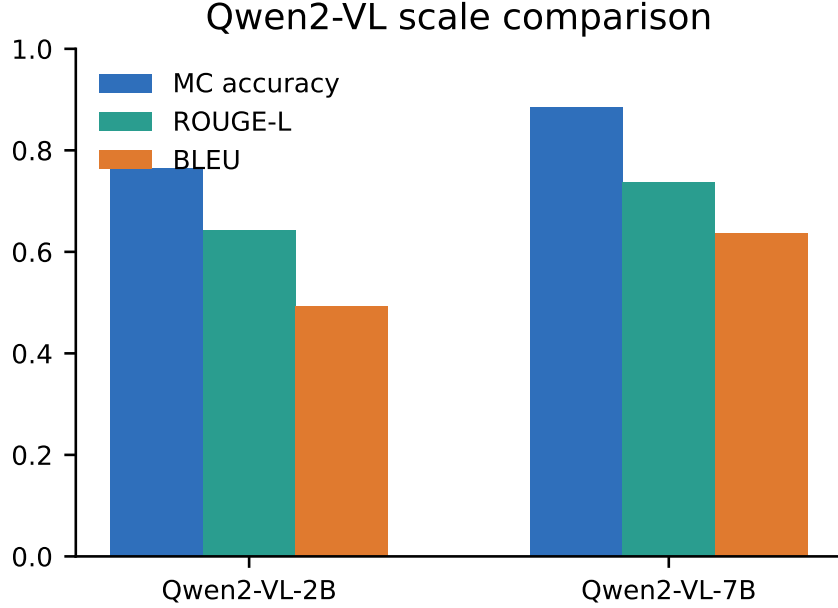


Figure 10: Scale comparison for the fixed ScienceQA explanation-aware setting. Moving from Qwen2-VL-2B to Qwen2-VL-7B improves both answer accuracy and rationale-overlap metrics.

4.6. Cross-domain transfer

Table 12 and Figure 11 lay out the train-domain against test-domain transfer matrix. Every column is scored in its target domain’s own metric, so the only safe comparisons run down a column, not across a row. Where the controls exist, the matrix keeps answer-only, rationale-generative, and explanation-aware sources separate. Treat it as an evaluation probe rather than the official in-domain table; a diagonal cell can drift a little from the matching completed run when the transfer evaluator routes through a different prompt family. The probe matters because answer-only training often travels better than rationale-generative training. The ScienceQA answer-only adapter is the clearest case, reaching 0.805 on A-OKVQA, 0.645 on ChartQA, 0.703 on DocVQA, and 0.630 on VQAv2. Its rationale-generative and explanation-aware siblings clear the random baselines but fall off on ChartQA, DocVQA, and VQAv2. A-OKVQA sources split the honours: rationale-generative training wins DocVQA transfer in that group at 0.764, while answer-only wins VQAv2 at 0.700. DocVQA is strongest at home (0.910), and the document-trained adapter carries over to ChartQA too

450 (0.685). The pattern, in short: transfer depends on the target domain and
 451 the source objective together, not on domain alone.

Table 12: Cross-domain transfer matrix. Rows indicate the training domain and columns indicate the evaluation domain. Each column uses the target domain’s headline metric, so values are best compared within a column.

Trained on	ScienceQA	A-OKVQA	ChartQA	DocVQA	VQAv2
ScienceQA answer-only	0.800	0.805	0.645	0.703	0.630
ScienceQA rationale CE	0.780	0.760	0.450	0.559	0.300
ScienceQA explanation-aware	0.740	0.765	0.460	0.491	0.410
A-OKVQA answer-only	0.690	0.825	0.635	0.677	0.700
A-OKVQA rationale CE	0.665	0.790	0.645	0.764	0.600
A-OKVQA explanation-aware	0.680	0.795	0.615	0.763	0.640
ChartQA answer-only	0.420	0.415	0.660	0.831	0.680
DocVQA answer-only	0.450	0.435	0.685	0.910	0.660
VQAv2 answer-only	0.440	0.575	0.660	0.716	0.730

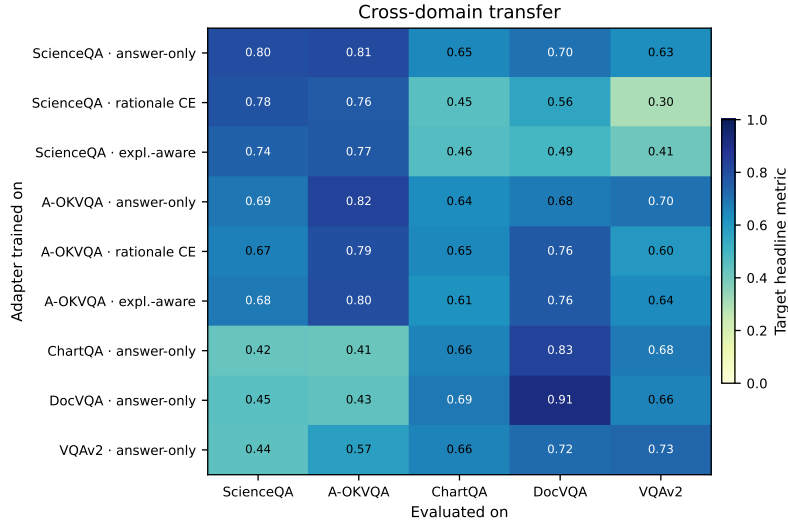


Figure 11: Cross-domain transfer heatmap. The strongest non-diagonal transfer is not always from the visually closest domain. Answer-only ScienceQA transfers strongly to A-OKVQA, while answer-only document and chart adapters transfer strongly to several open-ended targets.

452 5. Discussion

453 The results back the central idea, but only with caveats attached. They
454 also slot neatly into the arc the literature review traced: VLM alignment has
455 drifted from image–text representation matching toward generative instruc-
456 tion following, harder multimodal reasoning tests, and separate audits for
457 hallucination, critique, and evidence use. Objective choice clearly matters;
458 explanation supervision being automatically better than plain answer-only
459 fine-tuning does not follow. It can help, yes, but only when it is balanced with
460 care and held against a true answer-only control. On ScienceQA the best
461 fixed explanation-aware run is the low-answer-weight one, and the length-
462 aware control merely matches the rationale-generative control. The answer-
463 span-only setting is plainly worse on both accuracy and rationale quality.
464 A-OKVQA shows a small observed edge for explanation-aware training over
465 the generative control, but the capped-run uncertainty is wide enough that
466 we will not call it a firm win. And the fixed $\alpha = 0.50$ ScienceQA run is
467 actually worse than both the rationale-generative and answer-only controls.
468 Adding explanation tokens, by itself, does not make the model better. The
469 loss balance is what does the work.

470 The BLIP-2 contrastive result earns attention precisely because it is so
471 weakly positive on accuracy. We built the contrastive-enhanced arm specifi-
472 cally so the comparison could be run head-on. In the completed ScienceQA
473 run it sits just 0.005 above the BLIP-2 generative control, well inside the
474 approximate confidence intervals. Retrieval barely separates from the frozen
475 and random controls either. The read is that the auxiliary image–answer In-
476 foNCE term, at least in this small Q-Former-side setup, cannot forge a clearly
477 stronger image–answer representation. For this experiment set, generative
478 and explanation-aware supervision rest on firmer ground than contrastive
479 enhancement.

480 The domain results are the clearest argument against trusting a single
481 benchmark. ChartQA, DocVQA, VQAv2, A-OKVQA, and ScienceQA every
482 one improves in the selected runs. Yet the reading stays mixed, because the
483 domains are not all testing the same kind of supervision. ScienceQA and A-
484 OKVQA carry rationale-bearing targets; ChartQA, DocVQA, and VQAv2
485 fall back to answer-only training, because no gold rationales are there to
486 use. The strong DocVQA number, then, is capped extractive document
487 answering, nothing more, not evidence of faithful document rationales and
488 not a public-test DocVQA claim. Any stronger document-reasoning claim

489 would first need document-specific rationale supervision, or a more causal
490 OCR and layout faithfulness test.

491 The faithfulness numbers come with a warning attached. High mask-
492 ing drift proves the model is sensitive to visual change; it does not prove
493 the explanation is faithful. DocVQA and ChartQA drift hard because they
494 live on local visual evidence, and in this run both also improve under plain
495 answer-only fallback. ScienceQA and A-OKVQA post better answer and ra-
496 tionale scores, yet their drift stays small and refuses to climb consistently
497 with explanation-aware training. Better rationale text, in other words, is not
498 the same animal as more faithful visual grounding. So the masking diag-
499 nostic earns a place beside BLEU and ROUGE-L, never a place instead of
500 them.

501 Scale hands us the cleanest positive result of all. Holding the ScienceQA
502 explanation-aware objective fixed, Qwen2-VL-7B beats the 2B model on both
503 accuracy and rationale metrics. That does not retire the 2B experiments,
504 since the small model is exactly what made the wide sweep affordable in the
505 first place. What it does say is that some of the 2B weakness is probably
506 about capacity, not objective.

507 Transfer is mixed in the same way. Answer-only ScienceQA jumps cleanly
508 to A-OKVQA, while the rationale-generative and explanation-aware Sci-
509 enceQA adapters lose ground on the open-ended targets. A-OKVQA repeats
510 the shape: rationale-bearing objectives help on some targets, but answer-only
511 is the stronger bet for VQAv2. The answer-only ChartQA and DocVQA
512 adapters trade well with each other under the target metrics, a hint that
513 structured, text-heavy images share something useful when it comes to adap-
514 tation. The conclusion is the same one again: transfer rides on the target
515 domain and the source objective together.

516 5.1. Limitations

517 Compute is the first limitation, and the obvious one. A single-GPU bud-
518 get keeps the study realistic, but it also caps the seeds, the model sizes, and
519 the full-dataset runs we could attempt. Rationale availability is the second.
520 ScienceQA and A-OKVQA ship gold explanations; DocVQA, ChartQA, and
521 VQAv2 do not, so they cannot carry the same rationale supervision, and
522 any teacher-generated rationale experiment on them must be reported apart
523 from gold-rationale training. The metrics are the third. BLEU and ROUGE-
524 L can wave off a perfectly valid explanation just because the wording differs,
525 and while the masking metric gets closer to actual evidence use, it hinges on

526 which regions are masked and turns noisy on dense documents and charts.
 527 The protocol is the fourth: capped train and evaluation subsets, one seed
 528 per full configuration. That is what makes the work feasible and auditable,
 529 but it also means small gaps are not statistically settled. And finally, the
 530 execution logs are good enough to validate the outputs, yet not good enough
 531 for precise wall-clock or peak-memory benchmarking. Taken together, these
 532 are the lines the paper will not cross in what it claims.

533 5.2. Future directions

534 The first thing future work should do is rerun the strongest configura-
 535 tions with more seeds and larger evaluation splits. After that, swap in newer
 536 open multimodal backbones while keeping the objective-isolation protocol
 537 untouched. A third push is better rationale annotation for chart and doc-
 538 ument datasets, the very places where faithful explanations would pay off
 539 most. The faithfulness test itself should grow more causal, editing document
 540 text, chart marks, or image regions in controlled ways rather than mask-
 541 ing blindly. There is also a clean comparison waiting between supervised
 542 explanation-aware training and preference-based or reinforcement-learning
 543 methods that reward grounding directly. And before any of this ships, de-
 544 ployment work should track latency, memory, and calibration alongside ac-
 545 curacy and faithfulness.

546 6. Conclusion

547 This paper asked what the alignment objective alone does to vision-
 548 language reasoning under a fixed, resource-constrained fine-tuning bud-
 549 get. The comparison spanned zero-shot baselines, BLIP-2 generative and
 550 contrastive-enhanced alignment, and Qwen2-VL in answer-only, rationale-
 551 generative, and explanation-aware forms, across document, chart, science,
 552 commonsense, and natural-image benchmarks. Throughout, we mea-
 553 sured answer accuracy, rationale quality, and evidence sensitivity together
 554 rather than one at a time. What holds up best in the completed runs is
 555 Qwen2-VL task adaptation: rationale supervision applied with care where
 556 gold rationales exist, answer-only fallback where they do not. BLIP-2
 557 contrastive-enhanced training is here as a route we genuinely tested, but
 558 its gain over BLIP-2 generative training is small and its retrieval diagnostic
 559 is weak. All five datasets, ScienceQA, A-OKVQA, ChartQA, DocVQA,
 560 and VQAv2, improve under the selected runs. The faithfulness analysis

lands on a distinction worth keeping: better explanation text and stronger visual evidence use are related, not identical. Explanation-aware training lifts rationale text in some settings, yet it does not reliably raise masking-based evidence dependence. Scale is the cleanest win, with Qwen2-VL-7B strengthening the ScienceQA explanation-aware result, while the transfer matrix shows alignment staying stubbornly target-domain dependent. The honest summary is a careful one: supervised rationales can help resource-constrained VLMs, but only when answer-only controls, domain shift, and the evidence test are weighed together.

Declarations

Author responsibility. The authors wrote, reviewed, and edited the manuscript and take full responsibility for the final content, results, and claims.

Use of generative AI. Generative AI tools were used only for language editing, code assistance, and consistency checks. All scientific decisions, experiment selection, result interpretation, and final manuscript approval remain the responsibility of the authors.

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Data, code, and artifacts. The experiments use public datasets cited in the paper. The code, configuration files, generated tables, figures, execution logs, and claim-to-evidence artifacts are maintained with the project repository and can be made available with the submitted supplementary material.

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